apsara

Date: __/__/__

4/4/2026

Guest lecture

DAA

Big O Notation

•) No of Input ↑ Sourcing ↑ Time ↑

Real World Appl:

not just time, space

(i) Const Factor:

Two O(n) may have large computation time

(ii) Memory & Cache:

Way the data is stored matters

→ Reading Data Time > Processing Time → I/O Bottle Neck

Production Mindset

① Cloud Computing:

Serverless Architecture: Reach server only when you need it (acc)

Tradeoffs:

① E Commerce Platform: ways → precompounded recommendation (lots storage & comp. power)

(eg) Aggregate vendors (multiple) acc to inventory (pipeline)

② OTT

↳ Streaming quality v/s Int

(#) 3 layers: Algo → System → Scale

→ Making Sol^n Work

① System Integration ② Compliance & Standards ③ Scalability

④ Continuous Execution ⑤ Observation & Monitoring

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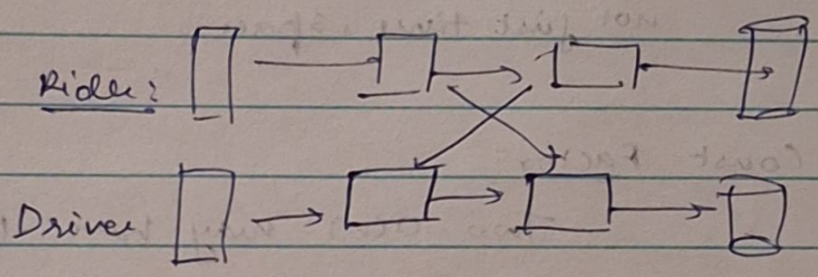
~~Ib Big System Goals~~

Recommendation System

Ride Hailing:



- low latency ~~Algo~~ Matching Algo
- compliment w real n well architectured design

- E-Commerce

Apache ~~Ea~~ → Pipeline
↳ streaming system

⇒ ML In production:

(i) Timely train models
→ Need a appropriate pipeline to stream

(ii) Scalability Data Pipelines Model Drift

Horizontal

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Vert
↓
add more processor & RAM

When Algo Meets the Cloud:

① Scalability changes when 'n' much

② Horizontal Scalability v/s Efficiency

③ Service changes the Cost Compatibility

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